

Individual Assignment 5

Alyssa Hagele

1. Set Up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# visualizing missing data
library(naniar)

# arranging plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
```

```

aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+"))

```

Class code

```

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),

```

```

        med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
        ostracod,
        copepod,
        hemiptera_corixidae_boatman,
        cladocera,
        nematode,
        diptera_ceratopogonidae,
        annelida_oligochaete,
        diptera_chironomid,
        annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5 liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>

```

```

# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm = TRUE),
  site_full = fct_rev(site_full))

# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
# filter to only include copepods
filter(taxon == "copepod") |>
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

# base layer
copepods <- ggplot(data = copepods,
  # x-axis: site
  # y-axis: mean abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
  shape = 21,
  width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
  fun = "median",
  size = 4,
  color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title

```

```

labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Copepods") +
# different theme
theme_bw()

# base layer
salinity <- ggplot(data = aq_ins_clean,
                  # x-axis: site
                  # y-axis: mean salinity
                  mapping = aes(x = site_full,
                               y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
            width = 0.1,
            shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,
            color = "red",
            size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
     y = "Median salinity (ppt)",
     title = "Site salinity") +
# different theme
theme_bw()

```

2. Problems

1. Visualizing differences across sites in major taxon abundance

a. Plan your figure

- x-axis: site_full
- y-axis: log_ave_lit
- geometry to represent average abundance/L: geom_jitter()
- geometry to represent medians: stat_summary()

b. Wrangle your data

```
#creating new object frm aq_ins_clean
ostracods <- aq_ins_clean |>
  #filter to only include ostracods
  filter(taxon == "ostracod") |>
  # log + 1 transform mean abundance per liter
  mutate(log_ave_lit = log(ave_lit + 1))

#display 10 random observations
ostracods |>
  slice_sample(n = 10)
```

```
# A tibble: 10 x 24
  date_site      date            site  taxon ave_lit med_ph med_sal med_do
  <chr>          <dtm>          <chr> <chr>  <dbl>  <dbl>  <dbl>  <dbl>
1 2023-06-06_NPB1 2023-06-06 00:00:00 NPB1 ostr~  0.533   7.48   NA     2.38
2 2022-02-23_NPB1 2022-02-23 00:00:00 NPB1 ostr~    0    NA    NA     NA
3 2023-03-03_NDC 2023-03-03 00:00:00 NDC  ostr~  5.73   7.45  0.688  5.64
4 2024-01-28_NPB 2024-01-28 00:00:00 NPB  ostr~  3.33   7.43  3.08   6.6
5 2023-11-12_NMC 2023-11-12 00:00:00 NMC  ostr~  0.533   NA    47    14.0
6 2023-05-05_NPB 2023-05-05 00:00:00 NPB  ostr~ 24.7    NA    NA    12.1
7 2023-05-04_NEC 2023-05-04 00:00:00 NEC  ostr~  0.667   8.05   NA    8.55
8 2022-05-11_NDC 2022-05-11 00:00:00 NDC  ostr~    0    9.11  0.398  9.4
9 2021-08-25_NPB 2021-08-25 00:00:00 NPB  ostr~ 581.    7.52  4.57   4.5
10 2023-11-11_NEC 2023-11-11 00:00:00 NEC  ostr~    0    NA   38.6   0.25

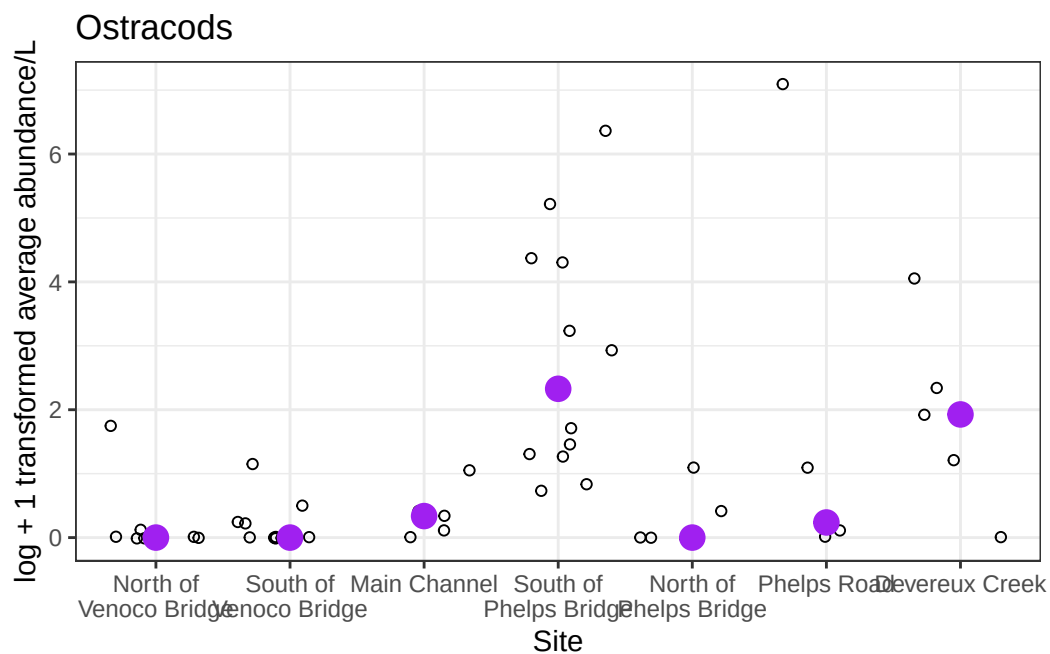
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c. Code your figure

```
# base layer
ostracods_plot <- ggplot(data = ostracods,
  # x-axis: site
  # y-axis: log + 1 transformed average abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
```

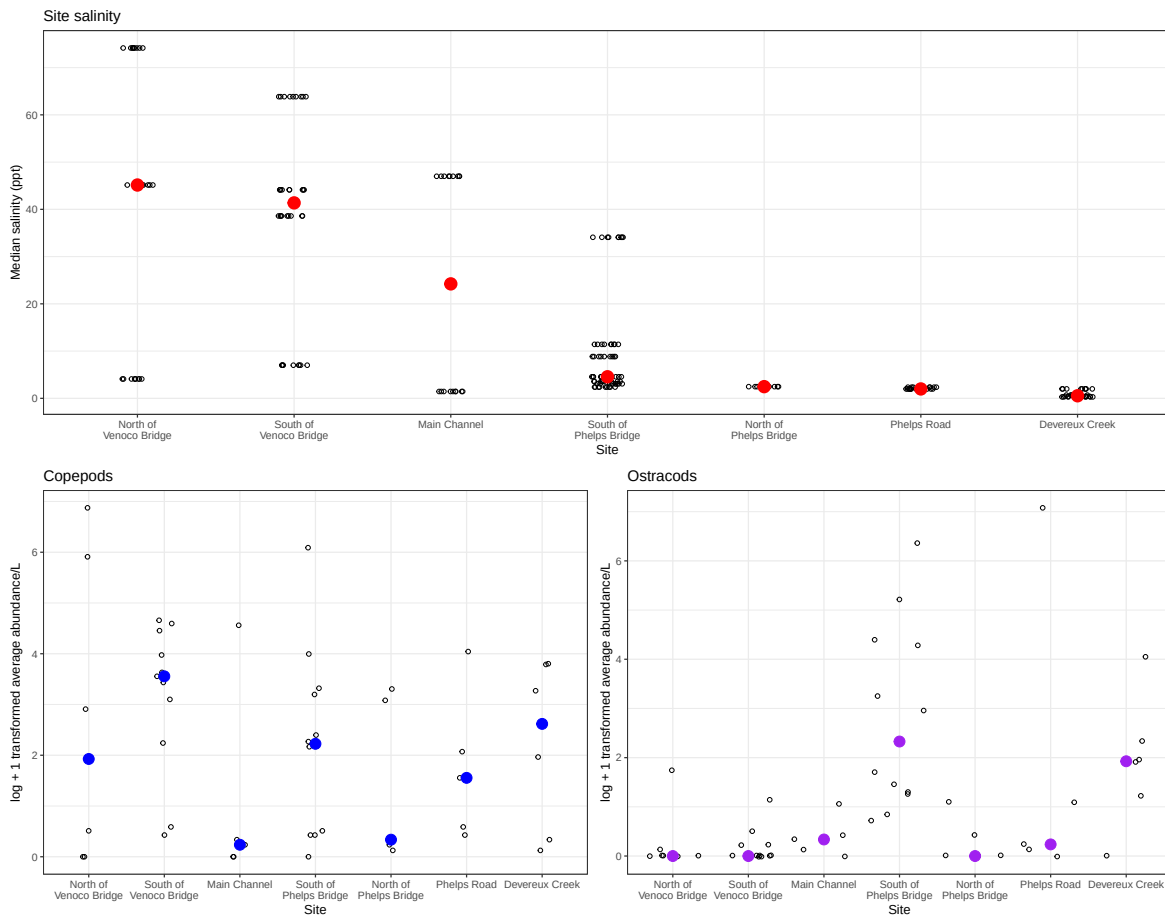
```
# first layer: points
geom_jitter(shape = 21) +
# second layer: median points
# changing size, color, shape
stat_summary(geom = "point",
             fun = "median",
             size = 4,
             color = "purple") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
     y = "log + 1 transformed average abundance/L",
     title = "Ostracods") +
# custom theme
theme_bw()

#showing plot
ostracods_plot
```



d. Create a multi panel figure

```
#using patchwork to create multipanel figure  
# *used chatgpt for help*  
salinity /  
  (copepods | ostracods_plot)
```



e. Interpret your figure

Differences in copepod and ostracod abundance vary across sites, with South of Venoco Bridge having the highest average copepod abundance with Devereux Creek following. This contrasts average ostracod abundance as South of Venoco Bridge has a very low average ostracod abundance, Devereux Creek has a similar amount of ostracods, and overall South of Phelps Bridge has the highest average ostracod abundance.

Ostracod abundance seems to not follow site differences in salinity / it appears to be influenced by other factors. For example, South of Phelps Bridge and Devereux Creek has the highest ostracod abundance, and also has low salinity (ppt), however so does Phelps Road and North of Phelps Bridge however they have low ostracod abundances.

2. Visualizing total abundance as a function of salinity

a. Plan your figure

- x-axis: med_sal
- y-axis: log_ave_lit
- color: taxon
- geometry: geom_point()
- facets: facet_wrap(~ taxon, scales = "free")

b. Wrangle your data

```
#creating new object from aq_ins_clean
aq_ins_new <- aq_ins_clean |>
  # create log transformed abundance column
  mutate(log_ave_lit = log(ave_lit + 1),
    #asked chat gpt for help to replace dashes with spaces in taxon names
    taxon = str_replace_all(taxon, "_", " "),
    #replacing taxons to be in title case
    taxon = str_to_title(taxon)) |>
  #selecting specific columns
  select(site_full,
    taxon,
    med_sal,
    ave_lit,
    log_ave_lit) |>
  #asked chat gpt for help reordering
  #ordering taxon levels by maximum abundance
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = max, .desc = TRUE, .na_rm = TRUE))

#showing 10 random observations from dataframe
aq_ins_new |>
  slice_sample(n = 10)
```

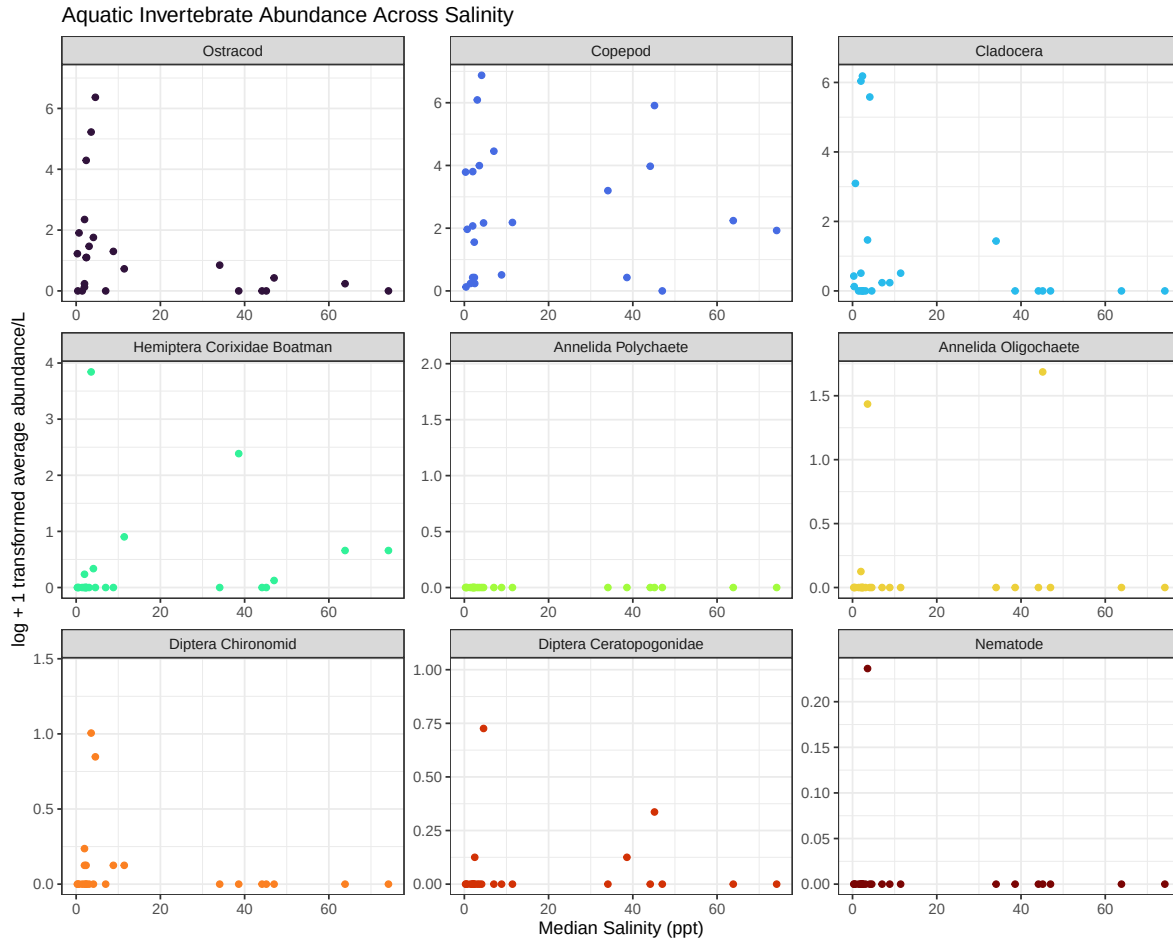
A tibble: 10 x 5

	site_full <fct>	taxon <fct>	med_sal <dbl>	ave_lit <dbl>	log_ave_lit <dbl>
1	Main Channel	Nematode	47	0	0
2	North of Phelps Bridge	Copepod	NA	26.3	3.31
3	Main Channel	Annelida Oligochaete	47	0	0
4	South of Phelps Bridge	Annelida Oligochaete	8.84	0	0
5	North of Phelps Bridge	Nematode	NA	0	0
6	South of Phelps Bridge	Annelida Polychaete	2.38	0	0
7	Main Channel	Cladocera	NA	0	0
8	Main Channel	Annelida Oligochaete	NA	0	0
9	Devereux Creek	Ostracod	0.398	0	0
10	Devereux Creek	Copepod	0.3	43.2	3.79

c. Code your figure

```
#base layer: ggplot
ggplot(data = aq_ins_new,
       #x axis is med_sal
       #y axis is log_ave_lit
       #color by taxon
       mapping = aes(x = med_sal,
                     y = log_ave_lit,
                     color = taxon)) +

#first layer: points
geom_point() +
#facet to create different panels by taxon
#free scales on both axis
facet_wrap(~taxon, scales = "free") +
#custom colors
scale_color_viridis_d(option = "H") +
#changing theme
theme_bw() +
#taking out legend
theme(legend.position = "none") +
#relabeling x and y axis
#relabeling title
labs(x = "Median Salinity (ppt)",
     y = "log + 1 transformed average abundance/L",
     title = "Aquatic Invertebrate Abundance Across Salinity")
```



d. Interpret your figure

Relationships between salinity and abundance varied across taxa. Ostracods, Copepods, Diptera Chironomids, and Cladoceras seemed to have relatively high abundances at low salinity, showing a possible negative relationship between salinity and abundance among these taxa. Several taxa such as Nematodes, Diptera Ceratopogonidae, Annelida Oligochaete, Annelida Polychaete, and Hemiptera Corixidae Boatmas didn't reveal a relationship as taxon abundance was low across salinity or points seemed sporadic compared to salinity.